Discrete Math W4

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October 13, 2026

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Hello

I am Diego

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Group Chat



Plan

1. Feedback
2. Theory Recap
3. Exercises
4. Questions?

Feedback

Theory Recap

Remark about predicate logic

When defining a predicate do it like this

$$P(x) = 1 \iff x \geq 1$$

$$P(x) = \begin{cases} 1 & x \geq 1 \\ 0 & \text{else} \end{cases}$$

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$$P(x) = \begin{cases} 1 & x \geq 1 \\ 0 & \text{else} \end{cases}$$

Also define functions and predicates separately

~~$P(f(y)) \iff f(y) \geq 2$~~

Where Are We?

- Basic Logic & Proofs
- ⇒ Sets & Relations
- Number Theory
- Algebra
- Logic

3. Sets & Relations

- Introduction
- Sets and Operations on Sets
- Relations
- Equivalence Relations
- Partial Order Relations
- Functions
- Countable and Uncountable Sets

The Set Concept

We define an elementhood predicate

$$E(x, y) = 1 \iff x \text{ is element of } y$$

Now we could postulate certain properties on this predicate.

In axiomatic set theory we would define a list of axioms and then build a theory out of them. In this course we use ZFC.

Set Equality

A set is completely specified by its elements.

Definition 3.2 (Axiom of extensionality)

$$A = B \overset{\text{def}}{\iff} \forall x(x \in A \leftrightarrow x \in B)$$

We postulate that if a is a set then $\{a\}$ is also a set.

Lemma 3.1

For any (sets) a and b, $\{a\} = \{b\} \implies a = b$

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$$\{1, 2, 23\} = \{1, 2, 2, 1, 23\} = \{23, 2, 1\} = \{342384728 - 342384727, 2, 23\}$$

Subsets

Definition 3.3

The set A is a subset of the set B, denoted $A \subseteq B$, if every element of A is also an element of B, i.e.,

$$A \subseteq B \overset{\text{def}}{\iff} \forall x \, (x \in A \to x \in B).$$

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Lemma 3.3

For any sets A, B, C.

$$A \subseteq B \land B \subseteq C \implies A \subseteq C$$

Union and Intersection

Definition 3.4

The *union* of two sets A and B is defined as

$$A \cup B \stackrel{\text{def}}{=} \{x \mid x \in A \vee x \in B\}$$

and their *intersection* is defined as

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Extension to a Family of Sets

Let $\mathcal{A}$ be a set of sets (a collection).

$$\bigcup \mathcal{A} \stackrel{\text{def}}{=} \{x \mid x \in A \text{ for some } A \in \mathcal{A}\}$$

$$\bigcap \mathcal{A} \stackrel{\text{def}}{=} \{x \mid x \in A \text{ for all } A \in \mathcal{A}\}$$

Some Examples

Let $\mathcal{A} = \{\{1,2\},\{2,3\},\{2,3,4\}\}$

$$\bigcup \mathcal{A} = \{1,2,3,4\} \qquad \text{and} \qquad \bigcap \mathcal{A} = \{2\}$$

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$$\bigcup \mathcal{A} = \{1,2,3,4\} \qquad \text{and} \qquad \bigcap \mathcal{A} = \{2\}$$

Let $A_n = \{n\}$ for $n \in \mathbb{N}$

$$\bigcup_{n=0}^{\infty} A_n = \mathbb{N} \qquad \text{and} \qquad \bigcap_{n=0}^{\infty} A_n = \emptyset$$

Difference

Definition 3.5

The difference of sets B and A, denoted $B \setminus A$ is the set of elements of B without those that are elements of A

$$B \setminus A \stackrel{\text{def}}{=} \{x \in B \mid x \notin A\}$$

In this course the complement is not defined so don't use it!

Some important laws

Theorem 3.4. *For any sets A, B, and C, the following laws hold:*

Idempotence:	$A \cap A = A$;
	$A \cup A = A$;
Commutativity:	$A \cap B = B \cap A$;
	$A \cup B = B \cup A$;
Associativity:	$A \cap (B \cap C) = (A \cap B) \cap C$;
	$A \cup (B \cup C) = (A \cup B) \cup C$;
Absorption:	$A \cap (A \cup B) = A$;
	$A \cup (A \cap B) = A$;
Distributivity:	$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$;
	$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$;
Consistency:	$A \subseteq B \iff A \cap B = A \iff A \cup B = B$.

All of these follow directly from the laws about propositional and predicate logic.

The Empty Set

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A set A is called **empty** if it contains no elements. So if

$$\forall x \neg (x \in A)$$

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Lemma 3.6

The empty set is a subset of every set.

$$\forall A (\emptyset \subseteq A)$$

Constructing Sets from the Empty Set

$$\emptyset \neq \{\emptyset\} \neq \{\{\emptyset\}\} \neq \cdots$$

Let

$$A = \{\emptyset\}, \quad B = \{\emptyset, \{\emptyset\}\}, \quad C = \{\emptyset\}, \quad D = \{B, E\}, \quad E = \{\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\}\}$$

Find which relations of the form $X = Y$, $X \subseteq Y$, or $X \in Y$ hold between these sets.

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Solution

- $C = A$ *(since both are* $\{\emptyset\}$*)*
- $A \subseteq B$ *and* $C \subseteq B$ *(elements of* A/C *are in* B*)*
- $A \subseteq E$*,* $B \subseteq E$*, and* $C \subseteq E$
- $B \in D$ *and* $E \in D$

$C = A \subseteq B \subseteq E \in D$

Construction of the Natural Numbers

We can define $\mathbb{N}$ from basic set theory using Von Neumann ordinals.[1]

$$\begin{aligned} 0 &= \{\} && = \emptyset, \\ 1 &= \{0\} && = \{\emptyset\}, \\ 2 &= \{0, 1\} && = \{\emptyset, \{\emptyset\}\}, \\ 3 &= \{0, 1, 2\} && = \{\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\}\}. \end{aligned}$$

The successor is defined as

$$s(n) \stackrel{\text{def}}{=} n \cup \{n\}$$

Addition can be defined as

$$m + 0 \stackrel{\text{def}}{=} m \quad \text{and} \quad m + s(n) \stackrel{\text{def}}{=} s(m + n)$$

[1] https://en.wikipedia.org/wiki/Set-theoretic_definition_of_natural_numbers

The Power Set

Definition 3.7

The power set of A, denoted $\mathcal{P}(A)$, is the set of all subsets of A

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\mathcal{P}(\emptyset) &= \{\emptyset\}, \\
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\end{aligned}$$

For finite sets $|\mathcal{P}(A)| = 2^{|A|}$

The Cartesian Product of Sets

Definition 3.8

The Cartesian product $A \times B$ of two sets A and B is the set of all ordered pairs with the first component from A and the second component from B

$$A \times B = \{(a, b) \mid a \in A \land b \in B\}$$

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Cartesian product of k sets

$$\bigtimes_{i=1}^{k} A_i = \{(a_1, \ldots, a_k) \mid a_i \in A_i \text{ for } 1 \leq i \leq k\}$$

The Cartesian Product

The cartesian product is not commutative or associative.

Not commutative: for $A = \{1, 2\}$ and $B = \{3\}$

$$A \times B = \{(1, 3), (2, 3)\} \neq \{(3, 1), (3, 2)\} = B \times A$$

Not associative: for $A = \{1\}$, $B = \{2\}$, $C = \{3\}$

$$(A \times B) \times C = \{((1, 2), 3)\} \neq \{(1, (2, 3))\} = A \times (B \times C)$$

since $((1, 2), 3) \neq (1, (2, 3))$.

Further, for any set A

$$A \times \emptyset = \emptyset \times A = \emptyset$$

Exercises

Some exam like tasks

Let $A = \{\emptyset, \{\emptyset\}, 1\}$. How many elements are in $\{A, 3, \emptyset\} \times \{A, A, 1\}$?

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Solution

$$\begin{aligned}\{A, 3, \emptyset\} \times \{A, A, 1\} &= \{A, 3, \emptyset\} \times \{A, 1\} \\ &= \{(A, A), (A, 1), (3, A), (3, 1), (\emptyset, A), (\emptyset, 1)\}\end{aligned}$$

6 Elements.

Compute the number of elements in the set
$|\mathcal{P}(\{\mathcal{P}(\{0\} \times \emptyset), \{\emptyset\}, 1\} \times \mathcal{P}(\{\emptyset\}))|$

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Solution

$$
\begin{aligned}
&|\mathcal{P}(\{\mathcal{P}(\{0\} \times \emptyset), \{\emptyset\}, 1\} \times \mathcal{P}(\{\emptyset\}))| \\
&= |\mathcal{P}(\{\{\emptyset\}, \{\emptyset\}, 1\} \times \mathcal{P}(\{\emptyset\}))| \\
&= |\mathcal{P}(\{\{\emptyset\}, 1\} \times \mathcal{P}(\{\emptyset\}))| \\
&= |\mathcal{P}(\{\{\emptyset\}, 1\} \times \{\emptyset, \{\emptyset\}\})| \\
&= 2^{|\{\{\emptyset\}, 1\} \times \{\emptyset, \{\emptyset\}\}|} \\
&= 2^{2 \cdot 2} \\
&= 2^4 \\
&= 16
\end{aligned}
$$

Let $A = \{\{\emptyset\}, \{\{\emptyset\}\}, \{\emptyset, \{\{\emptyset\}\}\}\}$. List all subsets of A that are not elements of A.

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Solution

$\emptyset$

$\{\{\{\emptyset\}\}\}$

$\{\{\emptyset, \{\{\emptyset\}\}\}\}$

$\{\{\emptyset\}, \{\{\emptyset\}\}\}$

$\{\{\emptyset\}, \{\emptyset, \{\{\emptyset\}\}\}\}$

$\{\{\{\emptyset\}\}, \{\emptyset, \{\{\emptyset\}\}\}\}$

A

Let A, B, and C be finite sets such that:

- $|(A \cup B) \times (B \cap C)| = 5$
- $A \cap B = \emptyset$
- $C \subseteq B$ and $|B| = 2$

Find the value of $|A| + |C|$.

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Find the value of $|A| + |C|$.

Solution

Since $C \subseteq B$, we have $B \cap C = C$, so

$$|A \cup B| \cdot |C| = 5$$

Because 5 is prime and $|C| \leq |B| = 2$, the only valid factor pair is $|C| = 1$ and $|A \cup B| = 5$ (since $|C| = 5$ violates $|C| \leq 2$).

Using $A \cap B = \emptyset$, we get $|A \cup B| = |A| + |B| \implies 5 = |A| + 2$, which gives $|A| = 3$.

$$|A| + |C| = 3 + 1 = \mathbf{4}$$

Let $|A| = n$.

Compute the number of subsets of $\mathcal{P}(\mathcal{P}(A) \setminus \emptyset) \setminus A$

Let $x \in A$ be fixed. Compute the number of subsets of $\mathcal{P}(A \setminus \{x\})$.

Let $|A| = n$.

Compute the number of subsets of $\mathcal{P}(\mathcal{P}(A) \setminus \emptyset) \setminus A$

Solution

Let $|A| = n$. The size of the inner set is

$$|\mathcal{P}(\mathcal{P}(A) \setminus \emptyset) \setminus A| = 2^{2^n - 1}$$

Since the number of subsets of a set of size k is 2^k, the total number of subsets is

$$2^{2^{2^n - 1}}$$

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Let $x \in A$ be fixed. Compute the number of subsets of $\mathcal{P}(A \setminus \{x\})$.

Solution

$$|\mathcal{P}(A \setminus \{x\})| = 2^{n-1}$$

So the number of subsets is

$$2^{2^{n-1}}$$

An important set identity

The **symmetric difference** of two sets A and B, denoted by $A \Delta B$, is defined as

$$A \Delta B = (A \setminus B) \cup (B \setminus A)$$

Prove that

$$A \cap (B \Delta C) = (A \cap B) \Delta (A \cap C)$$

If you are unsure about the validity, make a Venn diagram

Solution

Let $x \in A \cap (B \Delta C)$ be arbitrary. Then

$$
\begin{aligned}
& x \in A \cap (B \Delta C) \\
& \overset{\cdot}{\iff} x \in A \wedge x \in B \Delta C \\
& \overset{\cdot}{\iff} x \in A \wedge \big(x \in (B \setminus C) \cup (C \setminus B)\big) \\
& \overset{\cdot}{\iff} x \in A \wedge \big((x \in B \wedge x \notin C) \vee (x \in C \wedge x \notin B)\big) \\
& \overset{\cdot}{\iff} \big(x \in A \wedge x \in B \wedge x \notin C\big) \vee \big(x \in A \wedge x \in C \wedge x \notin B\big) \\
& \overset{\cdot}{\iff} \big(x \in A \wedge x \in B \wedge x \notin C \wedge x \in A\big) \vee \big(x \in A \wedge x \in C \wedge x \notin B \wedge x \in A\big) \\
& \overset{\cdot}{\iff} \big(x \in A \cap B \wedge x \notin (A \cap C)\big) \vee \big(x \in A \cap C \wedge x \notin (A \cap B)\big) \\
& \overset{\cdot}{\iff} x \in (A \cap B) \setminus (A \cap C) \vee x \in (A \cap C) \setminus (A \cap B) \\
& \overset{\cdot}{\iff} x \in \big((A \cap B) \setminus (A \cap C)\big) \cup \big((A \cap C) \setminus (A \cap B)\big) \\
& \overset{\cdot}{\iff} x \in (A \cap B) \Delta (A \cap C).
\end{aligned}
$$

Another Proof

$$A \times (B \setminus C) = (A \times B) \setminus (A \times C)$$

Solution

Let $(x, y) \in A \times (B \setminus C)$ be arbitrary. Then

$$\begin{aligned}
&(x, y) \in A \times (B \setminus C) \\
&\overset{\cdot}{\iff} x \in A \land y \in (B \setminus C) \\
&\overset{\cdot}{\iff} x \in A \land (y \in B \land y \notin C) \\
&\overset{\cdot}{\iff} (x \in A \land y \in B) \land y \notin C \\
&\overset{\cdot}{\iff} (x, y) \in A \times B \land y \notin C \\
&\overset{\cdot}{\iff} (x, y) \in A \times B \land (x, y) \notin A \times C \\
&\overset{\cdot}{\iff} (x, y) \in (A \times B) \setminus (A \times C).
\end{aligned}$$

Exercise 4.5 Relating Two Power Sets ($\star\,\star$)

Prove or disprove each of the following statements.

1. $\mathcal{P}(A \cap B) = \mathcal{P}(A) \cap \mathcal{P}(B)$ for any sets A and B.
2. $\mathcal{P}(A \cup B) = \mathcal{P}(A) \cup \mathcal{P}(B)$ for any sets A and B.
3. $A \subseteq B \iff \mathcal{P}(A) \subseteq \mathcal{P}(B)$ for any sets A and B.